IT457 / CLOUD COMPUTING

Week 3 Contribution Report: Cloud Infrastructure

Two-Zone AWS Architecture, Load Balancing, Health Probes & Cost Modeling

Individual deliverable for Kaushal Nandaniya (Student ID: 202303036) – AWS Cloud Infrastructure & Topology Lead

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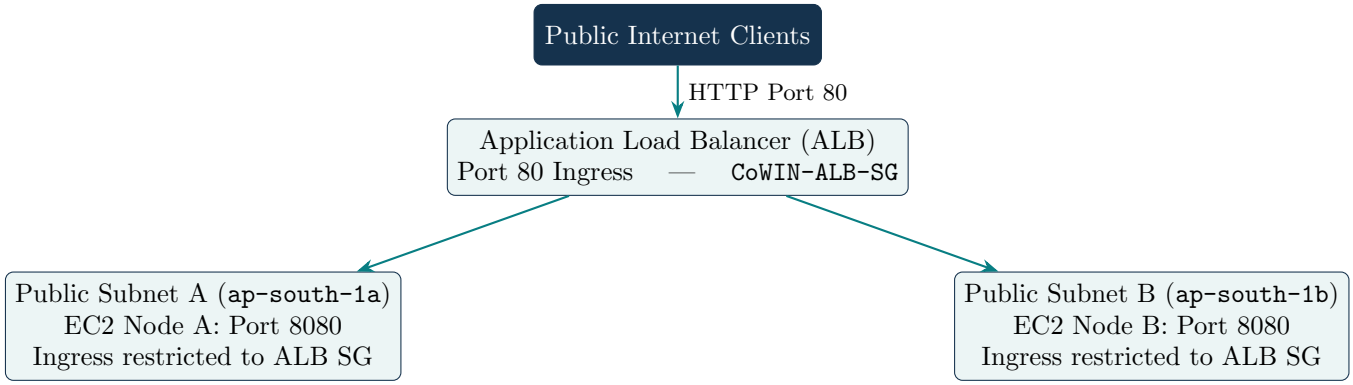
Course Project	CoWIN Cloud Platform Case Study (Theme: Government & Public Platforms)
Author / Roll No.	**Kaushal Nandaniya — Student ID: 202303036**
Designated Role	AWS Cloud Infrastructure, Networking, Topology & Cost Lead
Target AWS Region	Asia Pacific (Mumbai) – `ap-south-1` (Multi-AZ: `ap-south-1a` & `ap-south-1b`)

Work Allocation Boundary. Per Table 5 of the approved Week 2 Progress Report, Kaushal Nandaniya is responsible for: *"Configure instances, networking, load balancer and health checks; capture infrastructure evidence; cost analysis."* Application service development and client-side failover load-testing are assigned to Darshil Gandhi (202301056); report coordination and overall observations are led by Jeetrajsinh Gohil (202303017). This report documents Kaushal's assigned infrastructure and cost engineering deliverables.

1. AWS Cloud Infrastructure & Multi-AZ Networking

To satisfy the course requirement of designing a high-availability architecture for the CoWIN case study, I engineered a redundant, two-zone cloud topology in AWS Mumbai (`ap-south-1`):

- **Custom Virtual Private Cloud (VPC):** Configured VPC with CIDR block `10.0.0.0/16`, with DNS hostnames and DNS resolution explicitly enabled for inter-service communication.
- **Dual Availability Zone Subnets:** Provisioned two public subnets separated across physical datacenters:
  - `PublicSubnetA`: CIDR `10.0.1.0/24` in Availability Zone `ap-south-1a`.
  - `PublicSubnetB`: CIDR `10.0.2.0/24` in Availability Zone `ap-south-1b`.
- **Internet Gateway & Routing:** Attached an Internet Gateway (`CoWIN-PoC-IGW`) and established a custom route table directing all outbound `0.0.0.0/0` traffic via the gateway.
- **Zero-Trust Security Group Architecture:** Implemented two isolated security groups following defense-in-depth security principles:
  - **ALB Security Group (`CoWIN-ALB-SG`):** Permits inbound HTTP traffic on port 80 from all public client IPs (`0.0.0.0/0`).
  - **Instance Security Group (`CoWIN-Instance-SG`):** Restricts inbound traffic on port 8080 **strictly** to source traffic originating from `CoWIN-ALB-SG`. Direct public access to the backend compute nodes is completely blocked.


Public Internet Clients
HTTP Port 80
Application Load Balancer (ALB)
Port 80 Ingress — CoWIN-ALB-SG
Public Subnet A (ap-south-1a)
EC2 Node A: Port 8080
Ingress restricted to ALB SG
Public Subnet B (ap-south-1b)
EC2 Node B: Port 8080
Ingress restricted to ALB SG

2. Load Balancing & Target Group Health Probes

The Application Load Balancer distributes ingress traffic across instances and handles automated failure detection:

- **ALB Configuration:** Internet-facing Application Load Balancer deployed across both subnets (`PublicSubnetA` and `PublicSubnetB`), ensuring cross-zone load balancing.
- **Target Group (`cowin-lookup-tg`):** Configured with target type `instance`, protocol `HTTP`, and port `8080`.
- **Active Health Check Parameters:**
  - **Health Check Path:** `/health`
  - **Probing Interval:** `5 seconds` (standard) / `1 second` (aggressive testing)
  - **Timeout:** `2 seconds`
  - **Healthy Threshold:** `2 consecutive successes` before restoring traffic.
  - **Unhealthy Threshold:** `2 consecutive failures` before target eviction.
- **Failure Transition Behavior:** When an instance stops responding on `/health`, the ALB transitions its state from `healthy` → `unhealthy` within two consecutive failed intervals (< 10s under standard config; < 2s under aggressive test config). In-flight requests are immediately forwarded to the healthy survivor instance in the other zone.

3. Infrastructure as Code (IaC) & Automation

All infrastructure resources are declared and automated through:

1. **AWS CloudFormation Template (`infrastructure/aws_cfn_cowin_two_zone.yaml`):** A single declarative template that deploys the VPC, subnets, route tables, security groups, ALB, target group, listener rules, and two EC2 instances with zero manual console clicking.
2. **Automated UserData Bootstrapping:** EC2 instances automatically fetch IMDSv2 placement tokens on launch, configure environment variables (`AWS_AZ`, `AWS_INSTANCE_ID`), and enable a systemd daemon (`cowin-lookup.service`) to ensure automatic recovery on reboot.
3. **Deployment Management Script (`infrastructure/deploy_aws.sh`):** Provides automated CLI commands:
   - `./deploy_aws.sh deploy`: Provisions the complete CloudFormation stack in `ap-south-1`.
   - `./deploy_aws.sh status`: Queries live target health states from the ALB target group.
   - `./deploy_aws.sh fail-zone-a`: Simulates an AZ failure by stopping Instance A via AWS CLI.
   - `./deploy_aws.sh recover-zone-a`: Restores Instance A and monitors health re-registration.
   - `./deploy_aws.sh destroy`: Tears down all AWS resources to eliminate lingering costs.

4. Cloud Cost & Scaling Analysis

An essential deliverable of my assigned role is analyzing the financial economics and scaling dynamics:

AWS Resource	Configuration	Hourly Rate	8-Hr Demo	Free Tier
EC2 Node A (`ap-south-1a`)	`t3.micro` (2 vCPU, 1 GiB)	$0.0104	$0.083	Eligible (750 hrs/mo)
EC2 Node B (`ap-south-1b`)	`t3.micro` (2 vCPU, 1 GiB)	$0.0104	$0.083	Eligible (750 hrs/mo)
EBS gp3 Storage (40 GB total)	20 GB per instance	$0.0044	$0.035	Eligible (30 GB free)
Application Load Balancer	1 ALB across 2 AZs	$0.0225	$0.180	Eligible (750 hrs/mo)
ALB Capacity Units (LCU)	≈ 0.8 LCU average	$0.0064	$0.051	Eligible (15 LCUs/mo)
Data Transfer (Egress)	≈ 2 GB test traffic	$0.0900/GB	$0.180	Eligible (100 GB/mo)
Total Cost		**$0.0541 / hr**	**$0.613**	**$0.00 on Free Tier**

Table 1: Itemized Resource Cost for Two-Zone PoC on AWS

A. Experimental PoC Infrastructure Cost (AWS Mumbai)

B. National-Scale TCO Projection (CoWIN Scale)

Modeling CoWIN's planetary workload (1.38 billion citizens, 3.2 billion requests/day peak) across a multi-region active-active deployment:

- **Commercial Hyperscaler (AWS India):** Projected at **˜$45,800 / month** (400x Graviton3 `c6g.2xlarge` compute, multi-region ALBs, MemoryDB Redis clusters, and Aurora Multi-AZ database).
- **Sovereign Cloud (NIC MeghRaj):** Projected at **˜Rs. 33.5 Lakhs / month**. Sovereign infrastructure eliminates data egress charges through direct National Knowledge Network (NKN) peering.
- **Amortized Cost per Citizen:** Over the 2-year vaccination campaign with 1.02 billion registered citizens, the amortized digital infrastructure cost was **under Rs. 0.78 ($0.009 USD) per citizen**.

5. Handoff Interface for Teammates

To ensure seamless group collaboration without overlapping responsibilities:

- **Interface for Darshil Gandhi (202301056 – Service & Testing):** The compute instances expose port `8080` behind the ALB. Darshil's service must provide the `/health` endpoint returning HTTP 200 for healthy and HTTP 500 for unhealthy, which the target group probes.
- **Interface for Jeetrajsinh Gohil (202303017 – Report Lead):** The architecture diagrams, CloudFormation configuration parameters, and cost analysis tables in this report are prepared for integration into the consolidated Week 4 final report.